

Question1

December 9, 2025

1 DSCC 201/401 Final Project - Question1

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A. Copy the solar_data.tar.gz file from /public/bmort/R. Unpack the compressed file. Examine the data structure in the tables in each of the individual files.

```
In [57]: # Copy the solar_data.tar.gz file
system("cp /public/bmort/R/solar_data.tar.gz .")

# Unpack the compressed file
system("tar -xzf solar_data.tar.gz")

# List all files in the directory
files <- list.files("solar_data", full.name = TRUE)
files
cat("Number of data files:", length(files), "\n\n")

# Examine the data structure in the tables
lines <- system("head -30 solar_data/1997_DSD.txt", intern = TRUE)
cat(lines, sep = "\n")

lines <- system("head -30 solar_data/2025Q3_DSD.txt", intern = TRUE)
cat("\n", lines, sep = "\n")
```

```
1. 'solar_data/1997_DSD.txt' 2. 'solar_data/1998_DSD.txt' 3. 'solar_data/1999_DSD.txt'
4. 'solar_data/2000_DSD.txt' 5. 'solar_data/2001_DSD.txt' 6. 'solar_data/2002_DSD.txt'
7. 'solar_data/2003_DSD.txt' 8. 'solar_data/2004_DSD.txt' 9. 'solar_data/2005_DSD.txt'
10. 'solar_data/2006_DSD.txt' 11. 'solar_data/2007_DSD.txt' 12. 'solar_data/2008_DSD.txt'
13. 'solar_data/2009_DSD.txt' 14. 'solar_data/2010_DSD.txt' 15. 'solar_data/2011_DSD.txt'
16. 'solar_data/2012_DSD.txt' 17. 'solar_data/2013_DSD.txt' 18. 'solar_data/2014_DSD.txt'
19. 'solar_data/2015_DSD.txt' 20. 'solar_data/2016_DSD.txt' 21. 'solar_data/2017_DSD.txt'
22. 'solar_data/2017Q4_DSD.txt' 23. 'solar_data/2018_DSD.txt' 24. 'solar_data/2018Q1_DSD.txt'
25. 'solar_data/2018Q2_DSD.txt' 26. 'solar_data/2018Q3_DSD.txt' 27. 'solar_data/2018Q4_DSD.txt'
28. 'solar_data/2019Q1_DSD.txt' 29. 'solar_data/2019Q2_DSD.txt'
30. 'solar_data/2019Q3_DSD.txt' 31. 'solar_data/2019Q4_DSD.txt' 32. 'solar_data/2020Q1_DSD.txt'
33. 'solar_data/2020Q2_DSD.txt' 34. 'solar_data/2020Q3_DSD.txt'
35. 'solar_data/2020Q4_DSD.txt' 36. 'solar_data/2021Q1_DSD.txt' 37. 'solar_data/2021Q2_DSD.txt'
38. 'solar_data/2021Q3_DSD.txt' 39. 'solar_data/2021Q4_DSD.txt'
```

40. 'solar_data/2022Q1_DSD.txt' 41. 'solar_data/2022Q2_DSD.txt' 42. 'solar_data/2022Q3_DSD.txt' 43. 'solar_data/2022Q4_DSD.txt' 44. 'solar_data/2023Q1_DSD.txt' 45. 'solar_data/2023Q2_DSD.txt' 46. 'solar_data/2023Q3_DSD.txt' 47. 'solar_data/2023Q4_DSD.txt' 48. 'solar_data/2024Q1_DSD.txt' 49. 'solar_data/2024Q2_DSD.txt' 50. 'solar_data/2024Q3_DSD.txt' 51. 'solar_data/2024Q4_DSD.txt' 52. 'solar_data/2025Q1_DSD.txt' 53. 'solar_data/2025Q2_DSD.txt' 54. 'solar_data/2025Q3_DSD.txt' 55. 'solar_data/2025Q4_DSD.txt'

Number of data files: 55

:Product: Daily Solar Data 1997_DSD.txt

:Issued: 1900 UT 17 Dec 1998

#

Prepared by the U.S. Dept. of Commerce, NOAA, Space Environment Center.

Please send comments and suggestions to sec@sec.noaa.gov

#

1997 Daily Solar Data

#

Date	Radio		Sunspot		Stanford		GOES9		Flares						
	Flux	SESC	Area	New	Solar	X-Ray	X-Ray	Optical							
	10.7cm	Sunspot Number	10E-6 Hemis.	Regions	Mean Field	Bkgd Flux	C	M	X	S	1	2	3		
1997 01 01	72	0	0	0	*	A0.5	0	0	0	-1	-1	-1	-1		
1997 01 02	72	0	0	0	*	A0.5	0	0	0	-1	-1	-1	-1		
1997 01 03	73	0	0	0	3	A0.5	0	0	0	-1	-1	-1	-1		
1997 01 04	74	13	10	1	*	A0.6	0	0	0	-1	-1	-1	-1		
1997 01 05	74	15	20	0	5	A0.7	0	0	0	-1	-1	-1	-1		
1997 01 06	73	15	10	0	8	A1.0	0	0	0	-1	-1	-1	-1		
1997 01 07	73	12	0	0	*	A0.8	0	0	0	-1	-1	-1	-1		
1997 01 08	74	0	0	0	*	A0.8	0	0	0	-1	-1	-1	-1		
1997 01 09	74	12	20	1	*	A0.8	0	0	0	-1	-1	-1	-1		
1997 01 10	75	13	30	0	*	A0.8	0	0	0	-1	-1	-1	-1		
1997 01 11	74	12	20	0	*	A1.5	0	0	0	-1	-1	-1	-1		
1997 01 12	75	0	0	0	*	A1.0	0	0	0	-1	-1	-1	-1		
1997 01 13	75	0	0	0	*	A1.0	0	0	0	-1	-1	-1	-1		
1997 01 14	75	0	0	0	*	A0.9	0	0	0	-1	-1	-1	-1		
1997 01 15	76	14	30	1	*	A1.4	0	0	0	-1	-1	-1	-1		
1997 01 16	75	16	20	0	*	A1.4	0	0	0	-1	-1	-1	-1		
1997 01 17	74	12	0	0	-1	A0.8	0	0	0	-1	-1	-1	-1		
1997 01 18	75	34	0	2	*	A0.7	0	0	0	-1	-1	-1	-1		

:Product: Daily Solar Data quar_DSD.txt

:Issued: 2026 UT 07 Oct 2025

#

Prepared by the U.S. Dept. of Commerce, NOAA, Space Weather Prediction Center

Please send comments and suggestions to SWPC.Webmaster@noaa.gov

#

```

#               Quarterly Daily Solar Data
#
#               Sunspot           Stanford GOES15
#               Radio  SESC      Area      Solar  X-Ray  ----- Flares -----
#               Flux  Sunspot  10E-6   New    Mean  Bkgd      X-Ray      Optical
#               #   Date    10.7cm Number Hemis. Regions Field Flux      C  M  X  S  1  2  3
#-----
2025 07 01  126    144    300     2   -999     *    1  0  0  3  0  0  0
2025 07 02  130    148    315     1   -999     *    5  0  0  4  0  0  0
2025 07 03  125    118    300     0   -999     *    5  0  0  0  0  0  0
2025 07 04  119     87    360     0   -999     *    3  0  0  3  0  0  0
2025 07 05  117     82    220     0   -999     *    2  0  0  3  0  0  0
2025 07 06  118     81    230     1   -999     *    3  0  0  6  0  0  0
2025 07 07  118     96    290     1   -999     *    9  0  0  1  0  0  0
2025 07 08  115     88    290     0   -999     *    4  1  0  0  0  0  0
2025 07 09  120     83    290     2   -999     *    3  1  0  1  0  0  0
2025 07 10  129     72    340     0   -999     *   12  0  0  4  0  0  0
2025 07 11  132     98    490     3   -999     *   11  0  0 23  1  0  0
2025 07 12  139     93    700     0   -999     *   14  3  0 10  3  0  0
2025 07 13  128    102    710     1   -999     *   16  0  0  3  0  0  0
2025 07 14  128    139    545     2   -999     *   14  0  0  5  0  0  0
2025 07 15  139    146    585     2   -999     *    2  0  0  7  0  0  0
2025 07 16  152    182    900     2   -999     *    7  0  0 13  1  0  0
2025 07 17  156    178   1180     1   -999     *    3  0  0 14  0  0  0

```

These files are **whitespace-separated** ASCII text tables containing daily NOAA solar activity observations.

The file structure consists of three parts:

- (1) a metadata header at the top
- (2) multi-line ASCII-formatted column headings
- (3) daily data records beginning with the date in the form YYYY MM DD

Missing values are encoded as -1, -999, or *. All yearly and quarterly files almost share the same overall structure.

B. Preprocess the files in the Linux command line environment and then load the data from all years into a data frame in R. Produce a table of summary statistics on the data set. How do the ranges of the values in the columns with numerical data compare? Does each column of numerical data have similar magnitudes and ranges? Are there any outliers? Is there any missing data? Perform any data imputation or data modification as part of the data clean up process. Explain the reasoning for all actions taken.

In [81]: # 1. Preprocess the files in the Linux command line environment

```

# As required, I first used Linux commands (called through system() in R)
# to inspect and prepare the raw files:
# Used 'cp' to copy the compressed dataset into the working directory.
# Used 'tar -xvzf' to unpack the archive and extract all yearly/quarterly files.
# Used 'head' to examine the file structure.

```

```

# All these preprocessing steps have been finished in part A

# 2. Load the data from all years into a data frame in R

# Define a file-reading function
read_solar <- function(path) {
  # Skip the header line and detect the first sample
  raw <- readLines(path)
  data_start <- grep("^ *[0-9]{4}", raw)[1]

  df <- read.table(
    path,
    header = FALSE,
    comment.char = "#",
    fill = TRUE,
    skip = data_start - 1
  )

  # Add column names
  colnames(df) <- c(
    "Year", "Month", "Day",
    "Radio_Flux_10.7_cm",
    "SESC_Sunspot_Number",
    "Sunspot_Area",
    "New_Regions",
    "Stanford_Solar_Mean_Field",
    "GOES15_XRay_Bkgd_Flux",
    "XRay_C",
    "XRay_M",
    "XRay_X",
    "S",
    "Optical_1",
    "Optical_2",
    "Optical_3"
  )

  # Add Date column
  df$Date <- as.Date(
    with(df, sprintf("%04d-%02d-%02d", Year, Month, Day)),
    format = "%Y-%m-%d"
  )

  # Drop Year, Month, Day
  df <- subset(df, select = -c(Year, Month, Day))

  # Put Date as first column
  df <- df[, c("Date", setdiff(names(df), "Date"))]
}

```

```

# Columns that should be numeric (everything except Date + XRay_Bkgd_Flux string)
num_cols <- setdiff(names(df), c("Date", "GOES15_XRay_Bkgd_Flux"))
df[num_cols] <- lapply(df[num_cols], function(x) {
  if (is.factor(x)) x <- as.character(x)
  suppressWarnings(as.numeric(x))
})

df$GOES15_XRay_Bkgd_Flux <- as.character(df$GOES15_XRay_Bkgd_Flux)

return(df)
}

# Read all files as a dataframe
df_all <- do.call(rbind, lapply(files, read_solar))
cat("Dataset dimensions: ", nrow(df_all), "rows, ", ncol(df_all), "columns\n")

# Inspect the first few rows
print(head(df_all))

# 3. Produce a table of summary statistics on the dataset
summary(df_all)

# Check for duplicate dates
cat("# Duplicate dates", sum(duplicated(df_all$Date)), "\n")
# Remove fully duplicated rows (same Date and same values in all columns)
# caused by reading both yearly and quarterly DSD files
df_all <- df_all[!duplicated(df_all), ]
# Double check for duplicate dates
cat("# Duplicate dates after manipulation", sum(duplicated(df_all$Date)), "\n")
cat("# Dataset dimensions: ", dim(df_all)[1], "rows, ", dim(df_all)[2], "columns\n")

# Check missing values for Date and XRay_Bkgd
cat("# Missing values for Date: ", sum(is.na(df_all$Date)), "\n")
cat("# Missing values for XRay_Bkgd remarked as '*': ",
      sum(df_all$GOES15_XRay_Bkgd_Flux == "*", na.rm = TRUE), "\n")

make_summary <- function(df) {

  num_cols <- names(df)[sapply(df, is.numeric)]

  summary_table <- data.frame(
    Min      = sapply(df[num_cols], min, na.rm = TRUE),
    Q1       = sapply(df[num_cols], quantile, probs = 0.25, na.rm = TRUE),
    Median   = sapply(df[num_cols], median, na.rm = TRUE),
    Mean     = sapply(df[num_cols], function(x) round(mean(x, na.rm = TRUE), 2)),
    Q3       = sapply(df[num_cols], quantile, probs = 0.75, na.rm = TRUE),
    Max      = sapply(df[num_cols], max, na.rm = TRUE),
    Missing  = sapply(df[num_cols], function(x) sum(is.na(x)))
  )
}

```

```

    )

    return(summary_table)
}

# Summary for numeric variables
summary_df <- make_summary(df_all)
library(knitr)
kable(summary_df,
       caption = "Summary statistics for all numeric variables",
       digits = 2)

```

Dataset dimensions: 11000 rows, 14 columns

	Date	Radio_Flux_10.7_cm	SESC_Sunspot_Number	Sunspot_Area	New_Regions
1	1997-01-01	72	0	0	0
2	1997-01-02	72	0	0	0
3	1997-01-03	73	0	0	0
4	1997-01-04	74	13	10	1
5	1997-01-05	74	15	20	0
6	1997-01-06	73	15	10	0

	Stanford_Solar_Mean_Field	GOES15_XRay_Bkgd_Flux	XRay_C	XRay_M	XRay_X	S
1	NA	A0.5	0	0	0	-1
2	NA	A0.5	0	0	0	-1
3	3	A0.5	0	0	0	-1
4	NA	A0.6	0	0	0	-1
5	5	A0.7	0	0	0	-1
6	8	A1.0	0	0	0	-1

	Optical_1	Optical_2	Optical_3
1	-1	-1	-1
2	-1	-1	-1
3	-1	-1	-1
4	-1	-1	-1
5	-1	-1	-1
6	-1	-1	-1

	Date	Radio_Flux_10.7_cm	SESC_Sunspot_Number	Sunspot_Area
Min.	:1997-01-01	Min. : 64	Min. : 0.00	Min. : 0
1st Qu.:	:2004-07-12	1st Qu.: 74	1st Qu.: 13.00	1st Qu.: 20
Median	:2012-01-23	Median : 99	Median : 53.00	Median : 280
Mean	:2011-09-21	Mean : 112	Mean : 69.38	Mean : 1235
3rd Qu.:	:2018-09-02	3rd Qu.: 140	3rd Qu.: 112.00	3rd Qu.: 730
Max.	:2025-11-13	Max. : 343	Max. : 401.00	Max. : 8348570

	New_Regions	Stanford_Solar_Mean_Field	GOES15_XRay_Bkgd_Flux
Min.	:0.0000	Min. : -999.0	Length:11000
1st Qu.:	:0.0000	1st Qu.: -999.0	Class :character
Median	:0.0000	Median : -999.0	Mode :character

```

Mean    :0.5737   Mean    :-907.6
3rd Qu.:1.0000   3rd Qu.: -999.0
Max.    :6.0000   Max.    : 147.0
NA's    :36

```

```

      XRay_C      XRay_M      XRay_X      S
Min.   :-1.000   Min.   :-1.0000   Min.   :-1.000   Min.   :-1.000
1st Qu.: 0.000   1st Qu.: 0.0000   1st Qu.: 0.000   1st Qu.: 0.000
Median : 1.000   Median : 0.0000   Median : 0.000   Median : 1.000
Mean    : 2.861   Mean    : 0.3579   Mean    : 0.024   Mean    : 3.511
3rd Qu.: 5.000   3rd Qu.: 0.0000   3rd Qu.: 0.000   3rd Qu.: 5.000
Max.    :30.000   Max.    :22.0000   Max.    : 4.000   Max.    :55.000

```

```

      Optical_1      Optical_2      Optical_3
Min.   :-1.0000   Min.   :-1.00000   Min.   :-1.000000
1st Qu.: 0.0000   1st Qu.: 0.00000   1st Qu.: 0.000000
Median : 0.0000   Median : 0.00000   Median : 0.000000
Mean    : 0.2431   Mean    : 0.04218   Mean    :-0.002182
3rd Qu.: 0.0000   3rd Qu.: 0.00000   3rd Qu.: 0.000000
Max.    : 8.0000   Max.    : 4.00000   Max.    : 2.000000

```

```

# Duplicate dates 457
# Duplicate dates after manipulation 0
# Dataset dimensions: 10543 rows, 14 columns
# Missing values for Date: 0
# Missing values for XRay_Bkgd remarked as '*': 2108

```

	Min	Q1	Median	Mean	Q3	Max	Missing
Radio_Flux_10.7_cm	64	75	102	113.75	142	343	0
SESC_Sunspot_Number	0	14	57	72.09	115	401	0
Sunspot_Area	0	40	310	1287.68	760	8348570	0
New_Regions	0	0	0	0.59	1	6	0
Stanford_Solar_Mean_Field	-999	-999	-999	-903.67	-999	147	36
XRay_C	-1	0	1	2.98	5	30	0
XRay_M	-1	0	0	0.37	0	22	0
XRay_X	-1	0	0	0.03	0	4	0
S	-1	0	1	3.65	5	55	0
Optical_1	-1	0	0	0.25	0	8	0
Optical_2	-1	0	0	0.04	0	4	0
Optical_3	-1	0	0	0.00	0	2	0

How do the ranges of the values in the columns with numerical data compare? Does each

column of numerical data have similar magnitudes and ranges? Are there any outliers? Is there any missing data?

The numerical columns show **highly variable ranges and magnitudes**, and they do not operate on comparable scales. For example, `Radio_Flux_10.7_cm` ranges from 64 to 343, while `SESC_Sunspot_Number` spans 0 to 401. `Sunspot_Area` has an **extremely large upper bound of 8,348,570**, which is significantly larger than any other measurement and indicates the presence of **outliers**. The **flare-related variables** (`XRay_C`, `XRay_M`, `XRay_X`, `S`, `Optical_1/2/3`) have very **small ranges** (typically -1 to at most 55), reflecting their nature as sparse count data.

The column `Stanford_Solar_Mean_Field` is dominated by the sentinel value -999, with only a small number of valid measurements. This indicates **large missing data in this variable**. Besides that, missingness also comes from **non-numeric placeholder (*) in this column** which has been recognized when reading all files in R. **Several flare and optical variables** also use -1 as a **sentinel value for missing values**.

In the `GOES15_XRay_Bkgd_Flux` column, the placeholder value * indicates an invalid or unreported flux measurement and does not correspond to any physical X-Ray flux class. Therefore, it is treated as a **missing value**.

Overall, the numerical variables differ greatly in scale and distribution. Large physical measurements such as `Sunspot_Area` contain extreme high-end values (clear outliers), while flare count variables remain close to zero. Several columns contain missing data (-1, -999, *), which require cleaning and imputation before implementing ML models.

In [82]: # 4. Perform data imputation or data modification

```
# Based on summary, outlier lies in Sunspot_Area.
# Check outliers using the reasonable threshold
df_before_clean <- df_all
cat("Outlier values of Sunspot_Area:",
    df_before_clean$Sunspot_Area[df_before_clean$Sunspot_Area > 10000])

library(dplyr)
# Recode sentinel values to NA
sentinel_neg999_cols <- c("Stanford_Solar_Mean_Field")

flare_optical_cols <- c(
  "XRay_C", "XRay_M", "XRay_X", "S",
  "Optical_1", "Optical_2", "Optical_3"
)

df_all <- df_all %>%
  mutate(
    across(all_of(sentinel_neg999_cols),
      ~ ifelse(.x == -999, NA, .x)),
    across(all_of(flare_optical_cols),
      ~ ifelse(.x == -1, NA, .x))
  )

# Recode placeholder values to NA
df_all$GOES15_XRay_Bkgd_Flux[df_all$GOES15_XRay_Bkgd_Flux == "*"] <- NA
```



```

# Treat extreme Sunspot_Area as missing values
df_all <- df_all %>%
  mutate(
    Sunspot_Area = ifelse(Sunspot_Area > 10000, NA, Sunspot_Area)
  )

# Check the proportion of missing values after cleaning
missing_table <- data.frame(
  Variable = names(df_all),
  Missing_Percent_After_Cleaning = sprintf("%.2f%%", colMeans(is.na(df_all)) * 100)
)

kable(missing_table)

# Handle Missing values
library(tidyr)
df_clean <- df_all %>% drop_na(Sunspot_Area) #Remove outlier row
df_clean <- df_clean %>%
  select(-Stanford_Solar_Mean_Field) #Remove high-missing variable

low_missing_cols <- c("XRay_C", "XRay_M", "XRay_X", "S",
  "Optical_1", "Optical_2", "Optical_3")

# Imputation with median value
df_clean <- df_clean %>%
  mutate(across(all_of(low_missing_cols),
    ~ ifelse(is.na(.x), median(.x, na.rm = TRUE), .x)))

# Check the proportion of missing values after missing value imputation
missing_table <- data.frame(
  Variable = names(df_clean),
  Missing_Percent_After_Missing_Value_Imputation = sprintf(
    "%.2f%%",
    colMeans(is.na(df_clean)) * 100)
)

kable(missing_table)

# Check the distributions after missing value imputation
summary_df_clean <- make_summary(df_clean)
kable(summary_df_clean)

```

Outlier values of Sunspot_Area: 8348570

Variable	Missing_Percent_After_Cleaning
Date	0.00%
Radio_Flux_10.7_cm	0.00%
SESC_Sunspot_Number	0.00%
Sunspot_Area	0.01%
New_Regions	0.00%
Stanford_Solar_Mean_Field	90.52%
GOES15_XRay_Bkgd_Flux	19.99%
XRay_C	0.01%
XRay_M	0.01%
XRay_X	0.01%
S	0.85%
Optical_1	0.85%
Optical_2	0.85%
Optical_3	0.85%

Variable	Missing_Percent_After_Missing_Value_Imputation
Date	0.00%
Radio_Flux_10.7_cm	0.00%
SESC_Sunspot_Number	0.00%
Sunspot_Area	0.00%
New_Regions	0.00%
GOES15_XRay_Bkgd_Flux	19.99%
XRay_C	0.00%
XRay_M	0.00%
XRay_X	0.00%
S	0.00%
Optical_1	0.00%
Optical_2	0.00%
Optical_3	0.00%

	Min	Q1	Median	Mean	Q3	Max	Missing
Radio_Flux_10.7_cm	64	75	102	113.76	142	343	0
SESC_Sunspot_Number	0	14	57	72.10	115	401	0
Sunspot_Area	0	40	310	495.87	760	5690	0
New_Regions	0	0	0	0.59	1	6	0
XRay_C	0	0	1	2.98	5	30	0
XRay_M	0	0	0	0.37	0	22	0
XRay_X	0	0	0	0.03	0	4	0

S		0	0	1	3.67	5	55	0
Optical_1		0	0	0	0.26	0	8	0
Optical_2		0	0	0	0.05	0	4	0
Optical_3		0	0	0	0.01	0	2	0

Explain the reasoning for all cleaning actions taken.

We first recoded sentinel codes to proper NA. Measurements of the Stanford_Solar_Mean_Field use -999 to indicate **missing values**, and the flare-related variables use -1 to indicate unreported counts. These were converted to NA for consistency. We also identified one **extreme outlier** in Sunspot_Area (8,348,570), far beyond any physically possible value (normally <6,000), and recoded it as NA.

To handle missing values in the dataset, we applied **different strategies** depending on the nature and frequency of missingness in each variable. First, we **removed the row containing the extreme outlier** in Sunspot_Area (8,348,570), which was previously recoded as NA. This value is physically impossible, so deleting this single row prevents distortion of data. Next, we **excluded the Stanford_Solar_Mean_Field variable entirely**. This column contained more than **90% missing values**, making imputation unreliable and potentially misleading. Thus, removing the column is the most appropriate solution. For variables with very low missing rates (XRay_C, XRay_M, XRay_X, S, Optical_1, Optical_2, Optical_3), we performed **median imputation**. Using the median preserves the central tendency of the distribution without creating unrealistic values.

Since GOES15_XRay_Bkgd_Flux is a **categorical variable** and the missingness affects only this single field (with all other variables on the same row remaining valid), we retained those rows and **kept the NA values**.

C. Export your final clean data frame with all data combined from each of the years to a single csv file (solar_data_clean.csv).

```
In [45]: # Export the final clean dataframe as a csv file
write.csv(df_clean, "solar_data_clean.csv", row.names = FALSE)
list.files(pattern = "solar_data")

cat("#Final Dataset dimensions: ",
    dim(df_clean)[1], "rows, ",
    dim(df_clean)[2], "columns\n")
```

1. 'solar_data' 2. 'solar_data_clean.csv' 3. 'solar_data.tar.gz'

```
#Final Dataset dimensions: 10542 rows, 13 columns
```

D. Which day(s) had the highest value for the radio flux? Which day(s) had the lowest value for the radio flux?

```
In [77]: # Highest radio flux
max_flux <- max(df_clean$Radio_Flux_10.7_cm, na.rm = TRUE)
highest_days <- df_clean[df_clean$Radio_Flux_10.7_cm == max_flux,
    c("Date", "Radio_Flux_10.7_cm")]

kable(highest_days)
```

```
# Lowest radio flux
min_flux <- min(df_clean$Radio_Flux_10.7_cm, na.rm = TRUE)
lowest_days <- df_clean[df_clean$Radio_Flux_10.7_cm == min_flux,
                        c("Date", "Radio_Flux_10.7_cm")]

kable(lowest_days)
```

	Date	Radio_Flux_10.7_cm
10000	2023-02-17	343

	Date	Radio_Flux_10.7_cm
8785	2019-10-21	64

E. Which day(s) had the greatest number of solar flares? How many days had no solar flares?

```
In [78]: # Calculate the total flares = C + M + X + S + Optical_1 + Optical_2 + Optical_3
df_clean$total_flares <- df_clean$XRay_C +
                        df_clean$XRay_M +
                        df_clean$XRay_X +
                        df_clean$S +
                        df_clean$Optical_1 +
                        df_clean$Optical_2 +
                        df_clean$Optical_3

# Day(s) with the greatest number of solar flares
max_flares <- max(df_clean$total_flares, na.rm = TRUE)
days_with_max_flares <- df_clean[df_clean$total_flares == max_flares,
                                c("Date", "total_flares")]

kable(days_with_max_flares)

cat(
  "The day(s) with the greatest number of solar flares is/are: \n",
  paste(days_with_max_flares$Date, collapse = ", "),
  " with a total of ", max_flares, " flares.\n",
  sep = ""
)

# Number of days with zero solar flares
zero_flare_days <- sum(df_clean$total_flares == 0)
cat("Number of days without solar flares:", zero_flare_days)
```

	Date	total_flares
19994	2023-02-11	74

The day(s) with the greatest number of solar flares is/are:

2023-02-11 with a total of 74 flares.

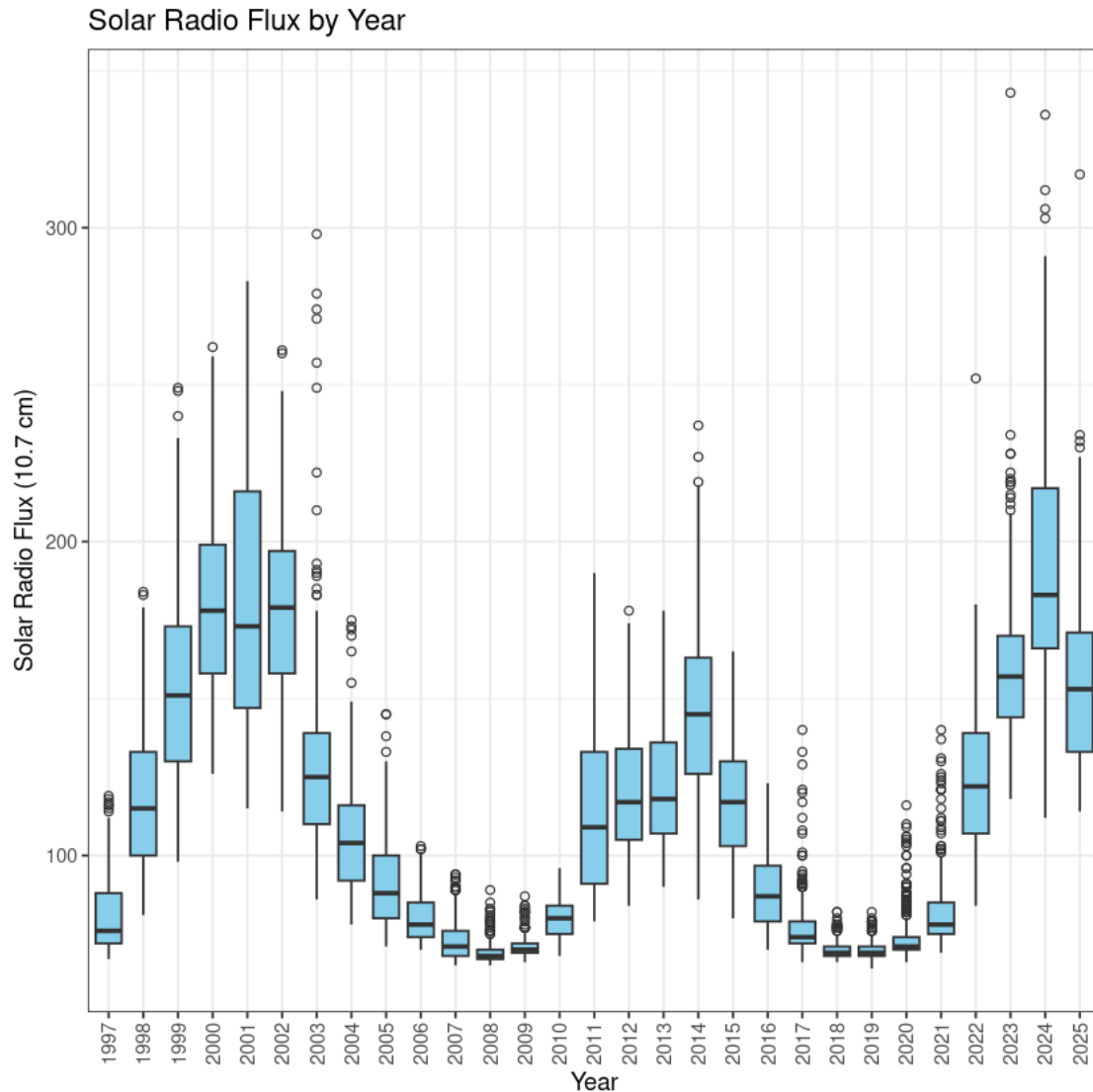
Number of days without solar flares: 3843

F. Create a single boxplot of the radio flux data. The y axis should be the solar radio flux and the x axis should be year.

```
In [48]: # Extract year value from date
df_clean$Year <- format(df_clean$Date, "%Y")

# Draw side-by-side boxplots
library(ggplot2)

ggplot(df_clean, aes(x = Year, y = Radio_Flux_10.7_cm)) +
  geom_boxplot(fill = "skyblue", outlier.shape = 1) +
  labs(
    title = "Solar Radio Flux by Year",
    x = "Year",
    y = "Solar Radio Flux (10.7 cm)"
  ) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5))
```



The yearly boxplot reveals **clear cyclical patterns** in the solar radio flux. **High flux values and greater variability** occur around **2000–2002, 2013–2015, and 2023–2025**, corresponding to solar cycle peaks. In contrast, the years **2008–2009 and 2018–2019** show **very low flux and minimal variability**, indicating solar minimum periods.

G. Partition the data set so that a random sample of 80% of the data will be used for training and 20% will be used for testing your machine learning model. The partition can be done randomly or the `createDataPartition()` function in R's `caret` library can be used.

```
In [49]: library(caret)
```

```
set.seed(123)
```

```
# Create the partition index (stratified sampling based on Radio Flux)
train_index <- createDataPartition(df_clean$Radio_Flux_10.7_cm,
```

```

p = 0.8,
list = FALSE)

# Split the data
train_data <- df_clean[train_index, ]
test_data <- df_clean[-train_index, ]

cat("Training set size:", nrow(train_data), "\n")
cat("Test set size:", nrow(test_data), "\n")

Training set size: 8435
Test set size: 2107

```

H. Generate a linear regression model to predict the solar flux from the sunspot number and the individual number of X-ray and optical solar flares.

```

In [50]: lm_model <- lm(
  Radio_Flux_10.7_cm ~
  SESC_Sunspot_Number +
  XRay_C + XRay_M + XRay_X +
  S + Optical_1 + Optical_2 + Optical_3,
  data = train_data
)

summary(lm_model)

```

Call:

```

lm(formula = Radio_Flux_10.7_cm ~ SESC_Sunspot_Number + XRay_C +
  XRay_M + XRay_X + S + Optical_1 + Optical_2 + Optical_3,
  data = train_data)

```

Residuals:

	Min	1Q	Median	3Q	Max
	-80.678	-7.751	-1.336	5.686	130.271

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	69.452597	0.255603	271.721	< 2e-16 ***
SESC_Sunspot_Number	0.535474	0.003632	147.442	< 2e-16 ***
XRay_C	1.374153	0.061525	22.335	< 2e-16 ***
XRay_M	4.596153	0.203657	22.568	< 2e-16 ***
XRay_X	9.351812	1.088890	8.588	< 2e-16 ***
S	-0.005983	0.044417	-0.135	0.89285
Optical_1	-1.030698	0.324131	-3.180	0.00148 **
Optical_2	0.601293	0.749049	0.803	0.42215
Optical_3	3.981631	2.196817	1.812	0.06995 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.77 on 8426 degrees of freedom

Multiple R-squared: 0.8731, Adjusted R-squared: 0.873

F-statistic: 7249 on 8 and 8426 DF, p-value: < 2.2e-16

I. Use the 20% of the data set aside for testing to determine the accuracy of your model. Choose an appropriate accuracy metric. How well does your model predict the solar radio flux from the sunspot number and number and type of solar flares? Comment on why the accuracy is good or poor.

```
In [51]: # Use test set for prediction
         pred <- predict(lm_model, newdata = test_data)

         # linear regression evaluation metrics: RMSE, MAE and R-squared
         rmse <- sqrt(mean((pred - test_data$Radio_Flux_10.7_cm)^2))
         mae  <- mean(abs(pred - test_data$Radio_Flux_10.7_cm))

         SSE <- sum((pred - test_data$Radio_Flux_10.7_cm)^2)
         SST <- sum((mean(test_data$Radio_Flux_10.7_cm) -
                        test_data$Radio_Flux_10.7_cm)^2)

         R2 <- 1 - SSE/SST

         cat("Test RMSE:", rmse, "\n")
         cat("Test MAE:", mae, "\n")
         cat("Test R-squared:", R2, "\n")
```

Test RMSE: 16.11226

Test MAE: 10.66703

Test R-squared: 0.8655497

RMSE, MAE and R-squared are the three **standard regression metrics** when evaluating the linear regression model's performance. On the test dataset, the model achieves:

- RMSE = 16.11

- MAE = 10.67

- R-squared = 0.87

Given that the solar radio flux ranges from 64 to 343 (a span of 279 units), the model's RMSE of 16.11 and MAE of 10.67 represent **approximately 6% and 4% of the total range** separately. This indicates that the linear regression model expresses very strong predictive accuracy in solar radio flux. An R-squared of 0.87 indicates that the model explains **87% of the variation in solar radio flux** using given explanatory variables.

This level of accuracy is quite strong for a multiple linear regression model.

J. What is the predicted radio flux for a day when 73 sunspots are observed on the sun and 2 solar flares with C-class energy levels and 1 solar flare with an M-class energy level occur? How confident are you in the answer? Explain your reasoning.


```

In [79]: # The given values of the explanatory variables
new_day <- data.frame(
  SESC_Sunspot_Number = 73,
  XRay_C = 2,
  XRay_M = 1,
  XRay_X = 0,
  S = 0,
  Optical_1 = 0,
  Optical_2 = 0,
  Optical_3 = 0
)

#Construct 95% prediction interval
# based on the given values of the explanatory variables
pred_CI <- predict(lm_model, newdata = new_day,
  interval = "prediction", level = 0.95)

kable(pred_CI)

```

fit	lwr	upr
115.8866	84.95992	146.8134

Given the specified explanatory variable values, the linear regression model predicts an **estimated solar radio flux of 115.9**, with a corresponding **95% prediction interval of 85.0 to 146.8**. I'm **95% confident** that the true radio flux of 73 sunspots, 2 C-class flares, and 1 M-class flare and no other flares lies between **85.0 and 146.8**.

This is a **reliable prediction** based on **our model's strong predictive ability** (RMSE = 16.11, MAE = 10.67, and R-squared = 0.87). Moreover, the values 73 sunspots, 2 C-class flares, and 1 M-class flare all fall well within the range of the training set, so the model is **interpolating rather than extrapolating**.